

# MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

## Tutorial 3

### Review: Lectures 6–7

- **First-order necessary condition.** If  $f : \mathcal{D} \rightarrow \mathbf{R}$  is differentiable and  $\mathcal{D} \subseteq \mathbf{R}^n$  is open, then

$$\mathbf{x}^* \text{ local optimizer} \implies \nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Such a point is called *stationary*.

- **Second-order necessary condition.** If  $f \in C^2(\mathcal{D})$ , then

$$\begin{aligned} \mathbf{x}^* \text{ local minimizer} &\implies \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq \mathbf{0}, \\ \mathbf{x}^* \text{ local maximizer} &\implies \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \preceq \mathbf{0}. \end{aligned}$$

- **Necessary  $\neq$  sufficient.**

$$\left. \begin{array}{l} \nabla f(\mathbf{x}^*) = \mathbf{0} \\ \nabla f(\mathbf{x}^*) = \mathbf{0}, \mathbf{H}_f(\mathbf{x}^*) \succeq \mathbf{0} \end{array} \right| \begin{array}{l} \text{candidate only} \\ \text{still not a minimum certificate} \end{array}$$

A stationary point may be a minimum, a maximum, or a saddle point.

- **Hessian test at a stationary point.**

$$\left. \begin{array}{l} \mathbf{H}_f \succ \mathbf{0} \\ \mathbf{H}_f \text{ indefinite} \end{array} \right| \begin{array}{l} \text{strict local min} \\ \text{saddle} \end{array} \quad \left. \begin{array}{l} \mathbf{H}_f \prec \mathbf{0} \\ \mathbf{H}_f \text{ singular semidefinite} \end{array} \right| \begin{array}{l} \text{strict local max} \\ \text{inconclusive} \end{array}$$

- **Unconstrained quadratic.** For  $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma$ ,  $\mathbf{Q} = \mathbf{Q}^T$ ,

$$\left. \begin{array}{l} \mathbf{Q} \not\succeq \mathbf{0} \text{ or } (\mathbf{Q} \succeq \mathbf{0}, \mathbf{c} \notin \text{Range}(\mathbf{Q})) \\ \mathbf{Q} \succeq \mathbf{0}, \mathbf{c} = \mathbf{Q} \mathbf{w} \end{array} \right| \begin{array}{l} \inf q = -\infty \\ \arg \min q = -\mathbf{w} + \text{Null}(\mathbf{Q}) \end{array}$$

- **Candidate search.**

$$\nabla f(\mathbf{x}) = \mathbf{0} \longrightarrow \text{necessary-condition filters} \longrightarrow \text{compare } f\text{-values.}$$

For a global conclusion, also prove a global lower bound; the stationary set may be infinite.

- **Domain warning.**  $\nabla f(\mathbf{x}^*) = \mathbf{0}$  is an interior condition. A boundary optimizer need not be stationary.

1. Let

$$\mathcal{D} = \{\mathbf{x} \in \mathbf{R}^3 : x_1^2 + x_2^2 + x_3^2 < 4\}$$

and consider minimizing on  $\mathcal{D}$  the function

$$f(\mathbf{x}) = \exp(x_1^2 - x_1) + x_2^2 - 4x_2 + 5x_3^2.$$

Does this problem have an optimal solution? Justify your answer using the first-order necessary condition.

2. Consider

$$q(x_1, x_2) = x_1^2 + 3x_1x_2 + 5x_2^2 + x_1 - 4x_2.$$

Write  $q$  as  $\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} + \mathbf{c}^T\mathbf{x}$ . Decide whether a global minimizer exists, whether it is unique, and find the minimizer and minimum value.

3. Consider

$$f(\mathbf{x}) = \ln(1 + x_1^2) + x_1^2 - 2x_1x_2 + 4x_2^2 + x_3^4 - 8x_3^3 + 16x_3^2, \quad \mathbf{x} \in \mathbf{R}^3.$$

(a) Find all stationary points.

(b) Determine all global minimizers of  $f$  over  $\mathbf{R}^3$ .

4. Consider

$$g(x_1, x_2, x_3) = x_1^6 + 5x_2^4 + x_3^2 - 3x_1x_2x_3^2.$$

(a) Verify that the origin is stationary and that  $\mathbf{H}_g(\mathbf{0})$  is positive semidefinite but singular.

(b) Classify the origin without relying on the Hessian test.

(c) Is this local conclusion also global?